

How Well Do LLMs Handle Bengali Healthcare NLP? A Cross-Task Evaluation

Anonymous ACL submission

Abstract

Large language models (LLMs) are increasingly used for healthcare-related natural language processing, but their capabilities in low-resource languages remain less understood. We present a unified benchmark of eight contemporary LLMs across five Bengali healthcare NLP tasks: depression detection, biomedical question answering, biomedical named entity recognition, health paraphrase generation, and consumer health-query summarization. The benchmark includes seven general-purpose multilingual models and one Bengali-specialized model, evaluated in a zero-shot setting with task-specific prompts and standardized metrics. Results show substantial variation across tasks, with no model consistently outperforming the others. Qwen 3.7 Plus achieves the highest accuracy on biomedical question answering and the highest F1 on depression detection, while Gemini 3.6 Flash performs best on biomedical NER. DeepSeek V4 Flash achieves the strongest results on health paraphrasing. Biomedical NER remains challenging, with substantially lower F1 than token-level accuracy. Biomedical question answering is particularly difficult for multi-answer questions, where accuracy drops sharply across all models. For summarization, model rankings vary across metrics, with Tiger LLM achieving the highest ROUGE-1 and Claude Haiku 4.5 the highest BERTScore. Overall, Bengali specialization does not consistently translate into stronger healthcare NLP performance. Our benchmark provides cross-task baselines for evaluating contemporary LLMs in Bengali healthcare settings.

1 Introduction

Large language models (LLMs) are increasingly used for healthcare-related natural language processing (NLP). They have been applied to tasks such as medical question answering, information extraction, and clinical text generation. Their

ability to perform multiple tasks without task-specific fine-tuning is especially promising for low-resource settings. However, their capabilities in low-resource languages and specialized domains remain less understood.

Bengali is an important example of such a setting. It is the sixth most spoken language in the world (Sani et al., 2026), yet remains low-resource in NLP. Recent work has introduced Bengali language models such as BanglaBERT (Bhattacharjee et al., 2022) and TigerLLM (Raihan and Zampieri, 2025). Researchers have also developed Bengali healthcare resources for tasks such as health-query summarization (Abrar et al., 2025) and mental health detection (Chowdhury et al., 2024). Despite this progress, most existing studies focus on a single task, dataset, or model. This makes it difficult to understand how modern LLMs perform across different Bengali healthcare tasks.

Healthcare NLP also involves diverse capabilities. Classification requires assigning predefined labels. Question answering requires selecting the correct answer or answers. Named entity recognition requires identifying medical entities and their boundaries. Generation tasks require producing new text while preserving the original meaning. A model that performs well on one type of task may therefore struggle on another.

To study these differences, we evaluate eight contemporary LLMs across five Bengali healthcare NLP tasks. The tasks cover depression detection, biomedical question answering, biomedical named entity recognition, health paraphrase generation, and consumer health-query summarization. Our evaluation uses a unified zero-shot framework. It includes seven general-purpose multilingual models and one Bengali-specialized model.

We investigate three questions:

- **RQ1:** How do contemporary LLMs perform across different Bengali healthcare NLP

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tasks?

- **RQ2:** Are model strengths consistent across classification, question answering, information extraction, and generation?
- **RQ3:** Does Bengali language specialization provide a consistent advantage in healthcare NLP?

Our results provide a cross-task evaluation of current LLM capabilities in Bengali healthcare NLP. We identify tasks where current models perform well and highlight persistent weaknesses that require further investigation. The benchmark also provides unified baselines for future research on Bengali healthcare language technologies.

2 Related Work

Recent work has established several Bengali healthcare NLP benchmarks. For mental health detection, Chowdhury et al. (2024) introduced the Bengali Social Media Depression Detection dataset (BSMDD) and evaluated transformer-based models and LLMs. Khan et al. (2023a) introduced BanglaCHQ-Summ for Bengali consumer health question summarization, while Khan et al. (2023b) introduced BanglaHealthNER for biomedical named entity recognition and showed that zero-shot GPT-3.5 substantially underperformed fine-tuned models.

Other benchmarks target generation and biomedical question answering. Aziz and Islam (2025) introduced BanglaHealth, a health-domain paraphrase dataset, while Sultana et al. (2025) introduced BanglaMedQA for Bengali biomedical multiple-choice question answering and investigated retrieval-augmented generation. Together, these studies provide important resources for Bengali healthcare NLP, but their evaluations remain largely task- or dataset-specific.

Our work differs by providing a unified zero-shot evaluation across five Bengali healthcare tasks. We evaluate eight contemporary LLMs, including seven general-purpose multilingual models and one Bengali-specialized model, covering depression detection, biomedical question answering, biomedical NER, paraphrase generation, and consumer health-query summarization. This enables us to examine both cross-task model consistency and whether Bengali specialization provides a consistent advantage.

3 Methodology

We evaluate eight contemporary LLMs across five Bengali healthcare NLP benchmarks. The evaluation follows a unified zero-shot protocol, with task-specific prompts and evaluation metrics selected according to each task.

3.1 Benchmarks and Tasks

We select five existing Bengali healthcare benchmarks covering classification, question answering, information extraction, and text generation. The Bengali Social Media Depression Detection dataset (BSMDD) is used for binary depression classification. Bengali biomedical question answering evaluates multiple-choice medical questions with single- and multi-answer instances. Bangla-HealthNER is used for token-level biomedical named entity recognition with seven entity types. BanglaHealth evaluates Bengali health paraphrase generation, while BanglaCHQ-Summ evaluates abstractive summarization of consumer health questions.

3.2 Language Models

We evaluate eight LLMs: Claude Haiku 4.5, DeepSeek V4 Flash, Gemini 3.6 Flash, GPT-5 Mini, Llama 4 Maverick, Mistral Large 3, Qwen 3.7 Plus, and Tiger LLM. Seven models are general-purpose multilingual LLMs, while Tiger LLM is Bengali-specialized. All models are evaluated without task-specific fine-tuning.

3.3 Evaluation Protocol

All models are evaluated in a zero-shot setting using task-specific system prompts. Prompts define the task objective and constrain the required output format. Classification and question answering prompts require structured predictions, while the BioNER prompt specifies the IOB2 tagging scheme and allowed entity types. Generation prompts require a single Bengali paraphrase or concise medical summary. The complete prompts and user templates are provided in Appendix B.

Model outputs are processed using task-specific parsing rules before evaluation. Structured responses are normalized to the expected format, while generated text is used after removing leading and trailing whitespace. Invalid structured responses are retried once; responses that remain invalid are treated as incorrect. Further implementation details and data preparation procedures are provided in Appendix D.

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Task	Metric	Claude Haiku 4.5	DeepSeek V4 Flash	Gemini 3.6 Flash	GPT-5 Mini	Llama 4 Maverick	Mistral Large 3	Qwen 3.7 Plus	Tiger LLM
CHQ-SUM	BLEU	32.99	40.87	35.57	34.67	40.17	38.78	35.49	40.82
	ROUGE-1	24.31	28.49	23.52	25.99	28.40	25.85	24.71	28.88
	ROUGE-2	7.92	10.34	6.83	8.38	9.70	8.03	7.95	10.27
	ROUGE-L	22.04	26.21	20.45	23.43	25.62	23.23	22.06	26.11
	METEOR	47.84	50.70	44.79	49.30	49.37	48.95	47.98	49.19
Biomedical QA	BERTScore	90.69	90.57	90.02	89.61	89.52	89.45	89.40	88.70
	Accuracy	83.50	86.96	90.85	75.86	83.60	83.50	91.95	53.52
Depression	Accuracy	78.59	82.44	82.70	79.65	76.35	71.50	82.95	71.30
	Precision	72.30	82.64	84.15	83.08	69.10	77.07	79.89	64.61
	Recall	92.34	82.87	82.70	79.65	95.60	71.50	88.11	94.31
	F1	81.10	82.76	83.38	81.07	80.22	69.94	83.80	76.69
Paraphrasing	BLEU	6.75	15.49	5.09	8.83	14.08	7.01	10.16	7.54
	ROUGE-1	33.49	45.55	29.27	35.75	43.12	33.14	39.73	34.15
	ROUGE-2	10.94	21.11	8.47	13.03	19.64	10.74	15.83	11.94
	ROUGE-L	30.67	41.75	26.42	33.50	39.05	29.77	36.32	32.16
	METEOR	27.36	39.22	23.02	29.24	37.14	26.55	34.13	27.56
BioNER	Accuracy	74.90	76.01	81.18	70.86	70.56	69.10	80.06	72.80
	Precision	16.32	21.25	33.09	10.51	12.70	9.60	30.85	9.78
	Recall	31.88	13.91	52.21	16.70	23.55	20.51	39.61	11.37
	F1	21.45	16.67	40.00	12.11	16.33	12.86	34.36	10.20

Table 1: Performance of the eight LLMs across five Bengali healthcare NLP tasks. Best results for each metric are shown in bold.

3.4 Evaluation Metrics

We use task-appropriate metrics and do not directly compare scores across different tasks. For depression detection, we report accuracy, precision, recall, and F1. For biomedical question answering, we report accuracy. For BioNER, we report token-level accuracy and weighted precision, recall, and F1, together with span-level results using seqeval (Nakayama, 2018). For health paraphrase generation, we use BLEU (Post, 2018), ROUGE-1, ROUGE-2, ROUGE-L (Lin, 2004), and METEOR (Banerjee and Lavie, 2005). For CHQ-SUM, we report ROUGE-1, ROUGE-2, ROUGE-L, and BERTScore (Zhang et al., 2020) using XLM-RoBERTa-base with Bengali as the selected language.

4 Results

Table 1 presents the performance of the eight LLMs across the five Bengali healthcare NLP tasks. Overall, the results show substantial variation across tasks. No model consistently achieves the best performance across all tasks. Models that perform well on classification and information extraction do not necessarily achieve the strongest results on generation tasks.

4.1 Cross-Task Performance

The classification results show relatively strong performance across most models. For depression

detection, Qwen 3.7 Plus and Gemini 3.6 Flash achieve the highest F1 scores, both at 0.83. Qwen 3.7 Plus also achieves the highest accuracy at 0.83, while Claude Haiku 4.5 shows high recall (0.92) but lower precision (72). Mistral Large 3 and Tiger LLM obtain the lowest F1 scores, at 70 and 77, respectively.

For biomedical question answering, Qwen 3.7 Plus achieves the highest accuracy (92), followed by Gemini 3.6 Flash (91) and Claude Haiku 4.5 (89). Overall, the results show substantial variation in medical question-answering performance across models. A detailed comparison of single-answer and multi-answer questions is provided in Appendix C.

4.2 Information Extraction and Generation

Biomedical NER is challenging for all models. Token-level accuracy is substantially higher than F1 for most models, with Gemini 3.6 Flash achieving the highest F1 (40), followed by Qwen 3.7 Plus (34). Several models obtain F1 scores below 20 despite accuracy above 70, showing that token-level accuracy alone can obscure poor entity recognition performance.

The generation tasks show a different model ranking. DeepSeek V4 Flash achieves the strongest results on health paraphrasing, with the highest BLEU (15) and ROUGE-1 (46), followed by Llama 4 Maverick. For consumer health-query summa-

rization, Tiger LLM achieves the highest ROUGE-1 (28.88), while Claude Haiku 4.5 obtains the highest BERTScore (90.69). The difference in model rankings across metrics highlights that lexical-overlap and embedding-based metrics capture different aspects of generated Bengali medical text.

5 Discussion

The evaluation shows that LLM performance is highly task-dependent. No model consistently performs best across all five Bengali healthcare benchmarks. Models that perform strongly on classification and information extraction do not necessarily achieve the best results on generation tasks. This pattern suggests that performance on a single Bengali healthcare benchmark is not sufficient to characterize a model’s overall capability in Bengali healthcare NLP.

The NER results highlight an important evaluation issue. Most models achieve substantially higher token-level accuracy than F1, indicating that accuracy alone can obscure poor entity recognition performance. This is particularly relevant for biomedical NER, where non-entity tokens constitute a large portion of the input. Entity-level metrics therefore provide a more informative view of model performance for this task.

The generation results also show differences across evaluation metrics. DeepSeek V4 Flash achieves the strongest lexical overlap on health paraphrasing, while the ranking for CHQ-SUM changes when BERTScore is considered. Claude Haiku 4.5 achieves the highest BERTScore, whereas Tiger LLM obtains the highest ROUGE scores. This suggests that lexical-overlap and embedding-based metrics capture different aspects of generated Bengali medical text and should be considered together.

Finally, Bengali specialization does not consistently translate into stronger performance. Tiger LLM performs competitively on summarization but performs substantially worse on several other tasks. Thus, specialization for Bengali alone does not appear sufficient to guarantee broad healthcare NLP performance. More targeted domain adaptation and evaluation are needed to determine how language specialization can translate into consistent gains across healthcare tasks.

6 Conclusion

We presented a unified zero-shot evaluation of eight contemporary LLMs across five Bengali healthcare NLP tasks. The results show substantial variation across tasks, with no single model consistently outperforming the others. General-purpose multilingual models frequently outperform the Bengali-specialized model, showing that Bengali specialization does not guarantee stronger healthcare NLP performance. Our findings demonstrate the importance of evaluating Bengali LLMs across multiple healthcare capabilities rather than relying on a single benchmark. Future work can extend this evaluation to additional healthcare tasks and datasets, as well as few-shot prompting, fine-tuning, human evaluation, and more detailed error analysis.

Limitations

This study evaluates models in a zero-shot setting only. Thus few-shot or fine-tuned performance may differ substantially. All evaluations rely on a single prompt template per task, so results may be sensitive to prompt phrasing. Dataset sizes for some tasks were reduced via stratified sampling, which may not fully capture the diversity of the original datasets. Finally, model behavior was evaluated at a single point in time, and API-hosted models may be updated or deprecated by providers, affecting reproducibility.

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Dataset	Task Type	System Prompt	User Prompt Template
BSMDD	Depression Classification (Binary)	Classify the given Bengali social media text into the following categories: 1 = Depressive (hopelessness, suicidal, self-harm, clear depression). 0 = Non-depressive (stress/anger alone does not equal depression). Output EXACTLY 0 or 1. No explanations. Nothing else.	Text: {text} Label:
Bangla Medical QA	Multiple-Choice QA	You are answering a multiple-choice question. Output only the correct option letter(s). If there is one correct answer, output exactly one of A, B, C, or D. If there are multiple correct answers, output the letters in alphabetical order separated by commas (e.g., A,B, B,D, or A,B,C). Output nothing else. Never give blank output.	Question: {question} A) {opt_a} B) {opt_b} C) {opt_c} D) {opt_d} Answer:
BioNER	Token-Level NER (IOB2)	You are a Bengali Biomedical Named Entity Recognition (BioNER) model. Label each input token using the IOB2 scheme. Entity types and their tags: Symptom → B-Symptom, I-Symptom; Health Condition → B-Health_Condition, I-Health_Condition; Medicine → B-Medicine, I-Medicine; Dosage → B-Dosage, I-Dosage; Medical Procedure → B-Medical_Procedure, I-Medical_Procedure; Specialist → B-Specialist, I-Specialist; Age → B-Age, I-Age; Outside → 0. Rules: Output exactly one tag per input token, one per line. Preserve token order strictly. Use B- for the first token of an entity span. Use I- for continuation tokens of the same entity. Output 0 for non-entity tokens. Return ONLY the labels, nothing else.	Sentence: {space_joined_tokens} Assign one IOB2 label to each token below. Return ONLY the labels in order, one per line. Tokens: {one_token_per_line}
BanglaHealth	Paraphrase Generation	You are a Bengali paraphrase generation model. Given a Bengali sentence, generate a single paraphrased version of it. Rules: Preserve the original meaning exactly. Vary the vocabulary and sentence structure. Output ONLY the paraphrased Bengali sentence, nothing else. Do not include explanations, labels, or any extra text.	Paraphrase this Bengali sentence: {sentence}
BanglaCHQ-Summ	Abstractive Summarization	You are a Bengali medical question summarization model. Given a Bengali health-related question asked by a patient, generate a concise summary. Rules: Retain all medically relevant information required to answer the question accurately. Be as concise as possible without discarding essential information. Preserve symptom details, duration, medications mentioned, and the core question. Output ONLY the Bengali summary, nothing else. Do not include explanations, labels, or any extra text.	Summarize this Bengali health question: {question}

Table 2: Task-specific system prompts and user prompt templates used for each benchmark.

Model	Overall	Single-Answer	Multi-Answer	Valid
Claude Haiku 4.5	83.50	84.16	11.11	994
DeepSeek V4 Flash	86.96	87.87	0.00	675
Gemini 3.6 Flash	90.85	91.57	11.11	994
GPT-5 Mini	75.86	76.55	0.00	994
Llama 4 Maverick	83.60	84.37	0.00	994
Mistral Large 3	83.50	84.16	11.11	994
Qwen 3.7 Plus	91.95	92.80	0.00	981
Tiger LLM	53.52	54.01	0.00	994

Table 3: Biomedical QA accuracy by answer type. Best overall accuracy is shown in bold.

Task	Example Raw Model Output	Parsed Output	Normalization Rule
Depression De- tection	1	1	N/A
Biomedical QA	"A,C"	AC	Extract all letters belonging to {A, B, C, D} from the raw string, sort alphabetically, and concatenate without separator . If no valid letter is extracted, the response is marked incorrect.
BioNER	B-Medicine i-medicine 0 0	B-Medicine I-Medicine 0 0	Split output by newline; map each line to a valid IOB2 label via exact match, then case-insensitive lookup against the full label vocabulary. Lines matching no valid label are mapped to 0. If the output contains fewer lines than the number of input tokens, the sequence is right-padded with 0; excess lines are truncated.
Paraphrase Gen- eration	[Bengali paraphrase]	[Bengali paraphrase]	Empty responses are excluded from metric computation.
CHQ-SUM	[Bengali summary]	[Bengali summary]	Responses that are empty or produced under an API error are excluded from metric computation.

Table 4: Task-specific output parsing and normalization rules applied before evaluation.